Encoder-first extraction with tiered, blind verification for MDS form filling

Methodology and first results, 5 November 2026. Design by the project owner. Tests use gpt-5-nano (low reasoning effort), local encoders and open datasets; MDS 2.0 Sections I, O and P on MIMIC-IV pilot admissions.

1. Components: three neural models, the LLM only for relations

Job	Done by	Role on MDS data
Mention tagging, assertion, subject, timing	One clinical encoder (Clinical ModernBERT), read once, three heads	Every mention in the discharge summary carries negation, who it is about and timing. Heads are trained on open labelled data (NegEx/ConText) and applied to MIMIC unchanged.
Synonyms, item class	Bi-encoder candidates, cross-encoder rerank; the LLM decides class membership	Only membership in an MDS item class is needed. The LLM reads the item definition, so no drug lists, and answers are cached per normalised string.
Abbreviations	Encoder detects; candidate senses; encoder picks; low margin goes to the LLM	Expansion is the hard part; a within-record consistency feature is preferred to a one-sense-per-note rule.
Dates	Encoder finds the expression and its anchor; code does the calendar arithmetic; LLM only for compositional cases	Windows are measured back from the assessment reference date (ARD). MIMIC notes mask dates, so record timestamps carry most dating.
Relations (event frames)	LLM	Medication (action, given / ordered / planned), treatment (status, setting), diagnosis (status, links to treatments).

2. Two levels of slot filling

- **Level 1:** fill event frames from the notes, encoders first and the LLM for gaps. One medication frame feeds many N items.
- **Level 2:** answer each MDS item as a query over the frames and their time intervals; an active diagnosis is a documented diagnosis plus a linked treatment event inside the window. The frame schema is designed by hand; the frame-to-item mapping is model-decided.

3. Verification: yes and no checked differently, then blind, in three tiers

- **Yes draft** (cited sentence): the patient received it (not held, not family); the date is inside the look-back before the ARD; the concept belongs to the item class. Only eMAR administrations and diagnosis codes count as direct record evidence; procedure codes and ICU events enter the evidence pack only.
- **No draft** (nothing found): a wider search with a lower tagging threshold, every medication or treatment mention, negated mentions and near misses. A no that the graph expects to be yes goes to that search first.
- **Three answers:** Yes, No, Unknown. No mention counts as No only if the stay covers the look-back window; otherwise it is Unknown.
- **Blind check:** every yes and every wide-search no goes to the LLM with the item definition and the evidence pack, never the draft, quoting support for Yes, No and Unknown. Agreement is accepted; a contradiction goes to the nurse with both evidence sets; an Unknown stays with its evidence attached.

Tier	Who	Settles
1	Encoders and code	Clear no answers: nothing linked, no record, no near miss, window covered.
2	Blind LLM	Every yes, unsure items, relative dates, class membership, and any disagreement that more evidence resolves; Unknown with evidence.
3	Nurse (labelling web UI)	Contradictions between the draft and the blind answer, shown with both evidence sets.

4. Results so far

Layer	Test	Result
Status heads	NegEx/ConText: 360 held-out mentions and NLI-checked rewrites; accuracy on original / negated sentences	Frozen encoder 0.81 / 0.25; fine-tuned shared encoder 0.86 / 0.45; LLM status arm 0.85 / 0.75; ConText rules 0.82 / 0.64
Dates	PMOA-TTS (silver labels), 20 cases, 284 events; within tolerance	Model gives the offset 0.49; anchor + amount, code arithmetic 0.41. 136 of 167 misses were the wrong direction.
Spans	Glaucoma lines / MACCROBAT, F1	LLM + verifier 0.88 / 0.52; NER + verifier 0.71 / 0.53; both must agree 0.71 / 0.50. No setup is best on both.
Blind vs shown draft	168 uncertain items (29 items, 15 admissions); precision against table proxies	Blind 0.78, shown the draft 0.59; shown the draft, the checker agreed with it on 63 items, blind on 52.
Tiers	15 admissions x 29 items, $0.14	With encoders also settling yes answers: Tier 1 58%, blind LLM 12%, nurse 30%. 91 of 174 no drafts fell in windows the stay does not cover.
Every yes checked blind	Same items, proxy labels	Precision of settled answers 0.42 to 0.87, recall 0.98; nurse share 30% to 62% when every Unknown is also escalated.
Agreement as confidence	CRF (Qwen3.8-27B), 80 dev notes	A value one of three methods gives is right 1-6% of the time; all three, 67%.
Path verifier	CRF test, 200 notes, Qwen; macro-F1	Graph + path verifier (no span verifier) 0.589 against the direct prompt 0.562.

Limits. Table proxy labels (codes and drug names over the whole stay) are noisy and share sources with record evidence, so precision figures are relative. Samples are small (15 admissions), and the record-source rule was derived on the same admissions it is scored on; a check on fresh admissions is running, as is the shared-versus-separate heads comparison. Heads need labelled training data, a departure from training-free. Abbreviation expansion and the i2b2 / n2c2 sets are untested (data access).